# Austin Moon

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## Education

**University of Waterloo** Waterloo, ON

*B.A.Sc. Mechatronics Engineering, Specialization in Artificial Intelligence* *Sep. 2024 – Expected June 2029*

## Technical Skills

**Languages**: Python, C/C++, SQL, VHDL, MATLAB
**AI/ML & Vision**: PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, ByteTrack, NumPy, pandas
**Embedded/Tools**: STM32, Arduino, PLC, I2C/SPI/UART, CAN bus, Git, Linux, Docker
**Design/Hardware**: SolidWorks, AutoCAD, Altium, KiCad, LTspice, PCB Fabrication, Soldering, Rapid Prototyping

## Experience

**Machine Learning Engineer** Sep 2026 – Present

*WAT.ai – OffBall XI* *Waterloo, ON*

- Working on reinforcement learning (Google Research Football) and graph-based models (PyTorch Geometric) to simulate counterfactual off-ball movement and decision-making from professional soccer tracking data
- Ran Google Research Football simulations to model a player's movement and presented findings to the team

**Tech Executive – AI/ML, Full-Stack Engineer** Sep 2026 – Present

*UW Association of Korean-Canadian Scientists and Engineers* *Waterloo, ON*

- Building an AI resume tailoring app that classifies job postings and scores resume bullets with Jev, rewrites them with LLM APIs using field specific prompts, and generates tailored resumes

**Embedded Software Developer** May 2025 – Aug 2026

*Waterloo Aerial Robotics Group* *Waterloo, ON*

- Developed STM32 firmware in C to drive a servo motor with PWM from SPI-based ADC input in real time
- Developed a C++ driver for a three-axis magnetometer on STM32, using I2C communication with DMA to read magnetic flux density and compute a real-time compass heading
- Extracted and parsed position, altitude, and speed from a u-blox M10 GPS receiver through an interrupt-driven UART NMEA stream, converting the data into navigation coordinates
- Implemented a low-pass filter in C for the ICM-42688 IMU within WARG's firmware, reducing noise in sensor data

**Mechatronics Engineering Intern** Sep 2025 – Dec 2025

*Spyder Controls Corporation* *Lacombe, AB*

- Led a UV chamber rebuild, wiring the control box, LED chamber, and cooling system with relays and sensors
- Built and repaired 15+ test jigs, diagnosing and fixing faulty circuitry and mechanical failures
- Flashed and validated firmware and circuits on 10+ products, systems, and test jigs over a CAN bus network
- Wrote a Python script for an Arduino Nano to flag out-of-tolerance (±5%), open, and shorted thermistors

**Mechanical Engineer** Feb 2025 – Nov 2025

*UW Orbital* *Waterloo, ON*

- Designed a modular 3-axis ADCS test platform in SolidWorks, gimballing ±30° in x/y with free z-axis rotation
- Engineered a custom bearing-supported gimbal on all 3 axes for low-friction, dynamic CubeSat orientation testing
- Implemented a stabilizing counterweight and reinforcement structure to support a minimum 4 kg load

**Engineering Assistant** Jan 2025 – Apr 2025

*University of Waterloo* *Waterloo, ON*

- Supervised the Rapid Prototyping Centre and Engineering Student Machine Shop, supporting 40+ students per day
- Set up and operated a new Rayjet R500 laser cutter, reviewing technical documentation and troubleshooting issues
- Processed 200+ laser cutting, 150+ 3D printing, and 50+ water jetting jobs using SolidWorks and AutoCAD
- Designed 2 drone attachments in SolidWorks in collaboration with the Waterloo Regional Police ERT

## Projects

**NFL Defensive Coverage Predictor** | *Python, PyTorch, YOLOv11, ByteTrack, OpenCV* | *GitHub*

- Trained a YOLOv11 detector on 10,000+ images (69% mAP50, 75% precision) with ByteTrack multi-object tracking
- Built automated camera calibration with Hough line transforms to map image pixels to real-world coordinates
- Engineered 339-dimensional features from 4.8M rows of tracking data to train a PyTorch MLP on 6 coverage types

**G-Helper Clone** | *Python, PyQt6, D-Bus, Linux* | *GitHub*

- Developed a PyQt6 system tray app to manage ASUS laptop hardware on Fedora Linux using Claude AI
- Tuned 50+ kernel and hardware parameters (CPU EPP, PCIe ASPM, PCI runtime PM), reducing idle power by 80%
- Built a debounced state machine to auto-configure CPU frequency, GPU DPM, and fan PWM on battery power